

1 What is Bias?

Three definitions (by field):

Field	Definition of bias
Psychology	Systematic deviation from rationality in judgment
Statistics	Systematic error in collection / analysis / interpretation of data
CS / ML	Systematic favoritism or discrimination toward certain groups/outcomes

How do we decide something is biased?

- **Social alignment** — does it clash with societal values? ('Women should cook while men work')
- **Stereotypes** — a widely held belief about some group/entity ('Black couples fight a lot')

When is bias a problem? Bias is **always there** (humans aren't perfect); it's only a problem **when it can have a negative impact**.

2 Demographics & Bias Perception

Demographics affect what counts as biased. 'That black man got into a fight' reads very differently in India vs the USA — 'black' isn't necessarily racist in the Indian context. Most existing bias **datasets are aligned to Western societies**, so they may miss or mislabel bias in other cultures.

3 Categories of Bias

Bias comes in many protected/social categories. Per the lecture, categories include: **Gender, Race, Profession, Religion, Age, Nationality, Socioeconomic status/Class, Physical appearance, Disability, Sexual orientation**. (Things like *scientific facts* or *currency exchange rates* are NOT bias categories.)

Assignment Q12: categories of bias = **Gender, Race, Profession** (NOT scientific facts, NOT currency exchange rates).

4 Types of Bias (know these by definition)

Type	Meaning / example
Reporting bias	Training data reflects extreme / non-representative reporting — people only report strong opinions (loved/hated a book), so subtle cases are under-represented
Historical bias	Data reflects past prejudice already existing in the world
Automation bias	Over-trusting the automated system's output over human judgment
In-group bias	Favoring members of one's own group
Selection / sampling bias	Data collected is not representative of all demographics
Latent design-choice bias	Hidden choices bias the data — e.g. using only English-language phrases

Assignment: the book-review-sentiment example (only strong opinions submitted) = **Reporting bias** (Q11). A 'latent' design choice that biases a dataset = **choosing to use only English-language phrases** (Q8). A **stereotype** = 'a widely held belief about some group' (Q10).

5 The Source of Bias — the ML pipeline

Bias can enter at **every stage** of the ML pipeline, not just one:

Stage	How bias enters
Collecting & labelling data	Is data representative of all demographics? Who are the annotators & what are their beliefs?
Training (metrics & objectives)	Training on biased data; using accuracy as a metric on unbalanced data hides poor minority performance
Deploying in production	What if users try to jailbreak the model? What if users perceive bias when there is none?
Users see an effect	(downstream consequence — <i>not itself a source of bias</i>)

Assignment Q6: bias is introduced at — **collecting & labelling data, training with chosen metrics/objectives, and deployment in production** (NOT 'users see an effect').

6 Measuring Bias — Benchmark Datasets

CrowS-Pairs (Crowdsourced Stereotype Pairs)

- Each item is a **pair of sentences**: one **more-stereotyping** and one **less-stereotyping** version
- Model is biased if it consistently prefers (assigns higher probability to) the more-stereotypical sentence
- Covers many bias categories (race, gender, religion, age, disability...)

StereoSet & related

- Measures stereotype vs anti-stereotype vs meaningless associations
- Trade-off: a model can be unbiased but useless, so score both **stereotype** and **language-modelling ability**

Indic / cultural context: Western-aligned datasets miss local bias; studies on themes like **Hatya (murder)** and **Dahej (dowry)** show prediction changes in Indian contexts.

Assignment Q7: each CrowS-Pairs item = **one stereotype sentence + one less-stereotype sentence**.

7 InSaAF — Fairness in the Indian Legal Landscape

InSaAF studies LLM fairness + accuracy for Indian legal tasks. It proposes a single combined metric — the **β -weighted Legal Safety Score (LSS)** — that **encapsulates both fairness and legal accuracy** (β trades off the two).

Why it matters: most fairness work is Western; InSaAF builds an **India-specific** pipeline & benchmark, showing generic LLMs can behave unfairly across Indian social groups in legal reasoning.

Assignment Q3: the InSaAF metric = **β -weighted Legal Safety Score** (not Demographic Parity, Perspective API, or Legal Grounding Score).

8 Debiasing with Contrastive Learning

Contrastive learning shapes an embedding space by pulling similar things together and pushing dissimilar things apart:

- If two sentences are **similar** → **minimise** the distance between their embeddings
- If two sentences are **different** → **maximise** the distance between their embeddings
- For debiasing: make representations **invariant** to the protected attribute so the model can't rely on it

Assignment Q9 (select all): correct contrastive-learning settings = **similar sentences** → **minimise distance** AND **different sentences** → **maximise distance**.

9 Real-World Bias Case Studies

Case	What happened / lesson
Microsoft Tay (2016)	Chatbot posted offensive tweets within hours due to malicious user intent (coordinated trolling) — deployment-time vulnerability
St George's Hospital (1988)	'A blot on the profession' — an admissions algorithm discriminated by gender & ethnicity (early real case)
Delphi (2021, Allen AI)	Ethical-advice model gave racist / problematic moral judgments
Predictive policing (PredPol)	Feedback-loop bias — trained on biased crime data → over-polices minority areas
AI image generators (Midjourney)	Ageism/sexism/classism — e.g. specialized/high-status roles shown as older men ; news analysts shown as younger
NYC AI hiring law (2023)	Regulation requiring bias audits of automated hiring tools

Assignment: Tay's issue = **posted offensive tweets due to malicious user intent** (Q4). Midjourney specialized-role bias = **only older men shown in specialized roles** (Q5).

10 Quick recap of exam-critical points

- **Bias definitions:** Psych = deviation from rationality; Stats = systematic error; ML = favoritism/discrimination
- **When a problem:** only when it causes negative impact (bias is always present)
- **Stereotype:** a widely held belief about a group
- **Categories:** gender, race, profession... (not scientific facts / exchange rates)
- **Sources:** data collection + training/metrics + deployment (not 'users see effect')
- **Reporting bias:** data over-represents strong/extreme cases
- **Latent design choice:** e.g. English-only phrases
- **CrowS-Pairs:** stereotype vs less-stereotype sentence pairs
- **InSaAF metric:** β -weighted Legal Safety Score (fairness + accuracy)
- **Contrastive learning:** similar→min distance, different→max distance
- **Tay:** malicious users; **Midjourney:** older men in specialized roles

Course activities this week: #Bias #Prompting (get a model to give a biased response — not gender), #Bias #CodedBias (watch the *Coded Bias* Netflix documentary), and the #Bias hands-on (run the supplied debiasing code yourself).